

JOSEPH GITAU CHEGE

Software Engineer — Backend Systems, Cloud Infrastructure & Reliability

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PROFESSIONAL SUMMARY

Software engineer with 3+ years building and operating production backend systems at scale. Proficient in **C#, Python, JavaScript/TypeScript, and C++** with a proven ability to own features end-to-end — from design through deployment and live-site support. Designed and shipped **notification and messaging capabilities** serving an institutional user base, with delivery across mobile app channels (push, in-app) alongside password and biometric authentication flows. Delivered 99.9% uptime on mission-critical services through disciplined incident response, monitoring, and post-incident improvement. Strong **Azure, Docker, and Kubernetes** background running globally available cloud-native services. Approaches problems with an AI-first, systems-thinking mindset and a focus on reliability, observability, and continuous modernisation.

CORE COMPETENCIES

Software Engineering	Cloud & Distributed Systems	Reliability & Live-Site Ops
C#, Python, JavaScript/TypeScript, C++ · Backend service & API development · Notification & messaging systems · Authentication flows · Database design (PostgreSQL, MySQL, MongoDB, Firebase) · CI/CD pipelines · Bash automation	Azure · Docker · Kubernetes · High-availability architecture · Scalable service design · Queue-based architectures · Proxmox · Resource & cost optimisation	Incident triage & escalation · Post-incident reviews · 99.9% uptime delivery · Monitoring & health checks · Enterprise-grade SLAs · SOPs & documentation · Cross-team collaboration

PROFESSIONAL EXPERIENCE

Engineering Team Lead & Head of IT · **Steadfast Academy**April 2025 – Present

Notification & Messaging Infrastructure

- Designed and built **notification delivery capabilities** within an in-house ERP system, enabling real-time push notifications, in-app alerts, and messaging features across mobile and web channels — directly analogous to high-scale notification service architectures.
- Implemented **authentication flows** including password-based login and on-device biometric verification, securing access to sensitive institutional data and demonstrating understanding of identity-layer concerns.
- Built a real-time **chat and messaging subsystem** within the ERP, handling message delivery, presence, and read-receipt patterns at institutional scale.

Platform & Backend Engineering

- Architected and developed the full-stack ERP system consolidating student records, attendance, finance, and communication workflows — replacing manual processes and providing a single source of truth across departments.
- Wrote Bash automation scripts and CI/CD pipelines that reduced deployment errors by 45% and reclaimed 15+ hours per week.
- Managed Azure cloud infrastructure end-to-end, optimising resource allocation and cutting cloud operating costs by 20%.
- Deployed and maintained containerised services using Docker and Kubernetes, improving system uptime by 50% and reducing latency by 30%.

Reliability, Incident Response & Live-Site Operations

- Delivered 99.9% uptime on mission-critical systems through proactive monitoring, scheduled maintenance, and disciplined, fully documented incident response — directly aligned with live-site/SLA-driven reliability engineering.

- Designed a high-availability storage architecture on TrueNAS supporting 10TB+ of institutional data, applying redundancy and failover patterns.
- Administered Ubuntu Server and Debian environments — service configuration, log analysis, and OS-level troubleshooting across production systems.

Infrastructure & Team Leadership

- Led a cross-functional engineering team and served as the primary technical point of contact between engineering and institutional stakeholders.
- Specced, built, and maintained the academy's on-premises server and network infrastructure from the ground up, including structured cabling, switch configuration, and asset tracking.

Software Engineering Intern · VisionFund Kenya (MicroFinance)Oct 2023 – Dec 2023

- Built internal automation tools in **C# and Node.js** and refactored MySQL database schemas, cutting query response times by 15% on critical financial reporting systems.
- Contributed to backend maintenance and debugging on live production financial systems, developing habits of careful, low-risk, well-documented changes within a regulated, high-accountability environment.

KEY PROJECTS

In-House ERP · Steadfast Academy2024 – 2025

- Designed and shipped a full-stack ERP application end-to-end — from requirements through deployment — consolidating student, attendance, finance, and communication records, with notification and messaging capabilities serving the entire institution.

BetterFarm — AgriTech Platform · Personal Project2024

- Lead architect for a cloud-native backend system on Azure; integrated real-time data pipelines and deployed scalable backend services using high-availability design principles.

Notification & Authentication Infrastructure · Steadfast Academy2025

- Designed and implemented real-time notification delivery (push, in-app) and authentication (password + biometric) subsystems within the ERP mobile and web platform, serving hundreds of daily active users.

TECHNICAL SKILLS

Languages	C# · Python · JavaScript/TypeScript · C++ · Bash · Kotlin
Backend & Data	PostgreSQL · MySQL · MongoDB · Firebase · API development · Schema design · Query optimisation · Notification systems · Messaging infrastructure · Authentication flows (password, biometric)
Cloud & DevOps	Azure (primary) · GCP (upskilling) · Docker · Kubernetes · Proxmox · CI/CD pipelines
Reliability & Ops	Linux (Ubuntu/Debian) · Incident triage & escalation · Monitoring · High-availability design · Enterprise SLAs · Documentation & SOPs
Networking & Hardware	LAN/WAN · VLAN · Patch panels · Firewall configuration · Server hardware · Structured cabling

EDUCATION

B.Sc. Computer Science (Coursework) · Umma University, Nairobi2021 – 2024

Key modules: Data Structures & Algorithms · Database Management Systems · Network Security · Operating Systems · Computer Architecture.